

Event-aligned conditional quantile trajectories of a biomarker before disease onset under left truncation and competing death

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Abstract

Biomarkers measured before the onset of a disease are observed only under delayed entry, loss to follow-up, competing death, and irregular visit times, and existing event-aligned models describe only their conditional *mean*. We introduce an event-aligned conditional *quantile* trajectory model: indexing time by both the visit age and the onset age, the conditional τ -quantile of the log-biomarker among individuals who develop disease is a smooth bivariate baseline surface plus a quantile-specific covariate effect β_τ . The estimand is identified from complete-event subjects without modeling the onset-time distribution and without a censoring model. Estimation solves an event-aligned check-function objective with a tensor-spline sieve for the baseline surface; we show that the resulting score is implicitly density-weighted and Neyman-orthogonal to the trajectory surface and the censoring nuisance, and we give a cross-fitted one-step estimator with $\sqrt{n}$ asymptotic normality and a clustered sandwich variance. Simulations confirm valid inference and recover non-parallel quantile trajectories; a comparison against the conditional-mean estimator of Sun et al. [2025] shows that the quantile model recovers distributional heterogeneity and induced non-proportional mean contrasts that a scalar mean coefficient collapses. We illustrate the method on cortical-thickness trajectories before amyloid positivity.

Keywords: conditional quantile regression; delayed entry; event alignment; Neyman orthogonality; sieve estimation; two time scales.

1 Introduction

Disease-relevant biomarkers often change in the years before a clinical landmark, and a central question is how that pre-onset trajectory depends on covariates. Cohorts that follow such biomarkers are subject to several sampling features at once: participants enter the study at staggered ages (delayed entry / left truncation), are lost to follow-up (right censoring), may die before developing the disease (competing risk), and are measured at irregular visit times. A useful descriptive target aligns each trajectory to the individual’s own onset age and asks how the biomarker behaves as a function of both the visit age u and the onset age t . Sun et al. [2025] formalized this two-time-scale view for the conditional *mean* of the marker and developed a semiparametric estimator under exactly these sampling complications.

The conditional mean, however, describes only the first moment. A covariate can leave the mean trajectory unchanged while shifting the *dispersion* of the marker, or it can act only in the *upper*

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tail of the distribution—changes that are clinically meaningful (e.g. a risk allele that accelerates decline only among the already-vulnerable) yet invisible to a mean model. More subtly, when a covariate’s effect varies across quantile levels, the *induced* mean contrast need not be proportional across (u, t) , so a single mean coefficient becomes a hard-to-interpret average. These are precisely the situations a conditional quantile model is designed to reveal.

We therefore model the entire family of conditional quantiles. Writing $L^0(u) = \log Y^0(u)$ for the log-biomarker at visit age u and T^0 for the onset age, we posit, for each $\tau \in (0, 1)$,

$$Q_\tau\{L^0(u) | T^0 = t, D^0 \geq t, \mathbf{X}^0\} = m_{\tau,0}(u, t) + \beta_{\tau,0}^\top \mathbf{X}^0, \quad (1)$$

where $m_{\tau,0}$ is a smooth bivariate baseline surface over the two time scales and $\beta_{\tau,0}$ is a quantile-specific covariate effect. The mean model of Sun et al. [2025] is the analogue at the level of the first moment; (1) recovers τ -varying covariate effects and, through them, distributional features the mean cannot express.

Contributions. (i) We define an event-aligned conditional quantile estimand (1) for the disease-developing subpopulation and give its identification from observed data under left truncation, right censoring, competing death, and irregular visits, conditioning on the observed onset age so that *no* onset-time model is required (Section 2). (ii) We estimate (1) by a complete-event sieve check-function objective and show that joint sieve quantile regression *implicitly* carries a density-weighted projection that makes the covariate score Neyman-orthogonal to the trajectory surface, and that the conditional-quantile restriction makes the score orthogonal to the censoring nuisance—so the default estimator needs no censoring model for identification (Section 3). (iii) We give a cross-fitted one-step estimator with $\sqrt{n}$ asymptotic normality and a clustered sandwich variance, and position the practical joint-sieve estimator relative to it (Section 4). (iv) Simulations establish valid finite-sample inference and, against the *actual* estimator of Sun et al. [2025], show that the quantile model recovers dispersion effects, non-proportional induced mean contrasts, and tail-specific effects that a scalar mean coefficient collapses (Section 5); an application to pre-amyloid cortical thickness illustrates the analysis (Section 6). Proofs, algorithms, and extensive sensitivity studies are in the Supplement.

2 Event-aligned quantile trajectory model

Target population and estimand. Time is age relative to a fixed origin. Let T^0 and D^0 be the ages at disease onset and death ($T^0 = \infty$ if disease-free at death); the target population is the disease-developing subpopulation $\{D^0 \geq T^0\}$. The estimand is the family $\{m_{\tau,0}(\cdot, \cdot), \beta_{\tau,0}\}_{\tau \in (0,1)}$ of (1): the conditional τ -quantile of $L^0(u) = \log Y^0(u)$ given onset age $T^0 = t$ and covariates $\mathbf{X}^0$, among individuals free of disease at the origin who develop disease. The covariate vector $\mathbf{X}^0$ contains no intercept and no deterministic function of (u, t) (these are absorbed into $m_{\tau,0}$, which would otherwise break identifiability of $\beta_{\tau,0}$). With the quantile score $\psi_\tau(r) = \tau - \mathbf{1}\{r \leq 0\}$, (1) is the conditional moment

$$\mathbb{E}\left[\psi_\tau\{L^0(u) - m_{\tau,0}(u, t) - \beta_{\tau,0}^\top \mathbf{X}^0\} | T^0 = t, D^0 \geq t, \mathbf{X}^0\right] = 0. \quad (2)$$

Observed data. Let A^0 be the age at study entry (delayed entry: A^0 may be negative). Subjects are retained only if $\max(A^0, 0) \leq \min(T^0, D^0)$ (left truncation). Let C be the time from entry to loss to follow-up; follow-up ends at age $A^0 + C$. Define $Z = \min(T^0, D^0, A^0 + C)$, $\Delta^T = \mathbf{1}\{T^0 \leq D^0, T^0 \leq A^0 + C\}$, $\Delta^D = \mathbf{1}\{D^0 < T^0, D^0 \leq A^0 + C\}$, $\Delta^C = \mathbf{1}\{A^0 + C < \min(T^0, D^0)\}$,

and $W = T^0 - A^0$ when $\Delta^T = 1$. The biomarker is observed at irregular pre-onset visit ages $U_{i\ell} \in (0, T_i)$, $\ell = 0, \dots, m_i$. Observed data are $\{(\mathbf{X}_i, A_i, Z_i, \Delta_i^T, \Delta_i^D, \{U_{i\ell}, L_{i\ell}\}_\ell)\}_{i=1}^n$.

Assumption 1 (Conditionally independent truncation). $A^0 \perp \{Y^0(\cdot), T^0, D^0\} \mid \mathbf{X}^0$, with positivity $\mathbb{P}(A^0 \leq u \mid T^0 = t, D^0 \geq t, \mathbf{X}^0 = \mathbf{x}) > c_A > 0$ on the estimable region.

Assumption 2 (Independent loss to follow-up). $C \perp \{Y^0(\cdot), T^0, D^0, N^{V,0}(\cdot)\} \mid (A^0, \mathbf{X}^0)$, where $N^{V,0}$ is the latent visit process.

Assumption 3 (Noninformative visits). $N^{V,0}(\cdot) \perp \{Y^0(\cdot), T^0, D^0\} \mid (A^0, \mathbf{X}^0)$.

These are process-level (not pointwise) independence statements; they are what the orthogonality argument of Section 4 uses, since they keep $\mathbb{P}(C \geq T - A \mid A, \mathbf{X}, \{U_\ell\}) = G_{C,0}(T - A \mid A, \mathbf{X})$ even after conditioning on the realized visit schedule.

Proposition 1 (Identification). Under Assumption 1, on the estimable region where $\mathbb{P}(A^0 \leq u \mid T^0 = t, D^0 \geq t, \mathbf{X}^0) > c_A$, the onset-conditioned quantile in the retained, complete-event sample equals the target $m_{\tau,0}(u, t) + \beta_{\tau,0}^\top \mathbf{X}^0$. Death before onset ($\Delta^D = 1$) places a subject outside the estimand and is handled by conditioning, not weighting; under Assumptions 2–3 neither a censoring model nor a visit-intensity weight is needed for identification.

The proof (Supplement A) verifies that conditioning on $(T^0 = t, \mathbf{X}^0)$ and the entry constraint removes the truncation selection, and that censoring and visits alter only the design density over the (u, t) plane, not the conditional quantile. This is the formal sense in which the model needs no nuisance model of the sampling mechanism to be identified; weights enter later only as an efficiency or design-standardization choice.

3 Estimation and orthogonal inference

3.1 Complete-event sieve quantile estimator

Let $\{B_k(u, t)\}_{k=1}^K$ be a tensor-product B-spline basis over the two time scales. Only complete-event subjects ($\Delta_i^T = 1$, for whom $T_i = Z_i$ and the basis $B(U_{i\ell}, T_i)$ is computable) and their pre-onset visits contribute. For a bounded positive observed-event weight $\omega(T, A, \mathbf{X})$,

$$(\hat{\gamma}_{\tau,\omega}, \hat{\beta}_{\tau,\omega}) = \arg \min_{\gamma, \beta} \frac{1}{n} \sum_{i: \Delta_i^T=1} \omega_i \sum_{\ell: 0 < U_{i\ell} < T_i} \rho_\tau \left(L_{i\ell} - B(U_{i\ell}, T_i)^\top \gamma - \mathbf{X}_i^\top \beta \right) + \frac{\lambda_n}{2} \gamma^\top P \gamma, \quad (3)$$

with $\rho_\tau(r) = r\{\tau - \mathbf{1}\{r < 0\}\}$ and $\hat{m}_{\tau,\omega} = B^\top \hat{\gamma}_{\tau,\omega}$. **The default is the complete-event weight** $\omega \equiv 1$, which estimates no censoring model. The inverse-probability-of-censoring weight $\omega = 1/\hat{G}_C(W \mid A, \mathbf{X})$, with $\hat{G}_C$ from a Cox model on the time-since-entry scale, is a design-standardization *sensitivity* variant that reweights the observed-onset sample to a prespecified design; by Proposition 1 both weights target the same $(m_{\tau,0}, \beta_{\tau,0})$. No onset-time model is used: we condition directly on the observed T_i .

3.2 Implicit density-weighted orthogonalization

The first-order conditions of (3) leave the $\mathbf{X}$ -block score $\mathbb{P}_n \omega \mathbf{X} \psi_\tau(\hat{\varepsilon}) = 0$ *uncentered*. The curvature of the objective, however, is the density-weighted Gram matrix with entries $f_{i\ell,\tau}(0) =$

$f_{\varepsilon_\tau|U,T,\mathbf{X}}(0|U_{i\ell}, T_i, \mathbf{X}_i)$. Profiling out γ from the linearized conditions yields, with $S_\lambda = D - C^\top A_\lambda^{-1} C$ the profiled information,

$$S_\lambda(\hat{\beta}_\tau - \beta_{\tau,0}) = \frac{1}{n} \sum_{i,\ell} \omega_i \{ \mathbf{X}_i - g_{\tau,K,\lambda}(U_{i\ell}, T_i) \} \psi_\tau(\varepsilon_{i\ell,\tau}) + \text{bias} + R_n, \quad (4)$$

where $g_{\tau,K,\lambda} = C^\top A_\lambda^{-1} B$. As $\lambda_n \rightarrow 0$ and the sieve grows, $g_{\tau,K,\lambda} \rightarrow g_{\tau,0} = d\mu_\tau/d\nu_\tau$, the density-weighted projection of $\mathbf{X}$ onto the trajectory surface (the Radon–Nikodym derivative of the visit occupation measures μ_τ, ν_τ weighted by $\omega f_\tau(0)$). The projected score $\{ \mathbf{X} - g_{\tau,0} \} \psi_\tau(\varepsilon)$ is therefore Neyman-orthogonal to m_τ : *joint sieve quantile regression performs the orthogonalization implicitly*. The two approximation biases in (4) (penalty and sieve) must be $o(n^{-1/2})$ for a centered limit; they bite first at extreme τ , where $f_\tau(0)$ is small.

3.3 Orthogonality to censoring, and a cross-fitted one-step

Plugging in $\hat{G}_C$ would normally add a first-order term; it does not here. With $\xi_{i,\tau} = \sum_\ell \{ \mathbf{X}_i - g_{\tau,0}(U_{i\ell}, T_i) \} \psi_\tau(\varepsilon_{i\ell,\tau})$, the conditional-quantile restriction and Assumption 3 give $\mathbb{E}[\xi_{i,\tau} | T, A, \mathbf{X}, D \geq T, \{U_\ell\}] = 0$, while the censoring perturbation is a function of $(T, A, \mathbf{X})$; by the tower property the censoring derivative vanishes (Lemma 1). The plug-in $\hat{G}_C$ then enters only through second-order products. As an explicit debiasing/validation device we estimate the sparsity $\hat{f}_\tau(0)$ by quantile spacings and the projection $\hat{g}_{\tau,\omega}$ by a density-weighted spline regression, and form a *cross-fitted* one-step estimator: with subject folds $I_1, \dots, I_K$, fit all nuisances and an initial $\hat{\beta}^{(-k)}$ off fold k and aggregate the held-out, bread-corrected scores,

$$\tilde{\beta}_{\tau,\omega}^{\text{CF}} = \sum_k \frac{n_k}{n} \left[\hat{\beta}_{\tau,\omega}^{(-k)} + \hat{S}_k^{-1} \hat{U}_{\tau,\omega,k} \right], \quad \hat{U}_{\tau,\omega,k} = \frac{1}{n_k} \sum_{i \in I_k} \hat{\phi}_i^{(-k)}. \quad (5)$$

Subject-level resampling preserves the within-subject clustering for inference; fold-discipline details are in Supplement B.

4 Large-sample theory

For a generic admissible weight ω (oracle nuisances), the subject-level influence function and information are

$$\phi_{i,\tau,\omega}^0 = \Delta_i^T \omega_i \sum_\ell \{ \mathbf{X}_i - g_{\tau,\omega}(U_{i\ell}, T_i) \} \psi_\tau(\varepsilon_{i\ell,\tau}), \quad S_{\tau,\omega} = \mathbb{E} \left[\mathbf{1}\{T \leq D\} r_\omega \sum_\ell f_\tau(0) \{ \mathbf{X} - g_{\tau,\omega} \}^{\otimes 2} \right],$$

with design weight $r_\omega = G_{C,0}\omega$. The complete-event default ($\omega \equiv 1$) uses $r_\omega = G_{C,0}$ and contains no $\hat{G}_C$; the IPCW instance ($\omega = 1/G_{C,0}$) gives $r_\omega = 1$.

Lemma 1 (Censoring orthogonality). *Under Assumptions 2–3 and (1), the pathwise derivative of the moment in the censoring nuisance vanishes for every admissible perturbation; hence $\hat{G}_C$ contributes no first-order term and enters only through second-order products $O_p(r_G(r_m + r_\beta))$. This is a local orthogonality property induced by the conditional moment restriction; it is not a double-robustness claim.*

Proposition 2 (Design-weight invariance). *For any bounded positive weight $w(T, A, \mathbf{X})$ with $0 < c_w \leq w \leq C_w$ on the estimable support and nonsingular weighted information, $w \equiv 1$ and $w = 1/G_{C,0}$ identify the same $(m_{\tau,0}, \beta_{\tau,0})$; w affects only efficiency and the finite-sieve pseudo-true value. The projection and information are weight-specific, $g_{\tau,w} = d\mu_w/d\nu_w$, so in general $g_{\tau,\text{CC}} \neq g_{\tau,\text{IPCW}}$.*

Theorem 1 (Cross-fitted one-step asymptotic normality). *Fix $\tau \in (0, 1)$ and an admissible weight ω . Assume the observed-data quantile restriction; $S_{\tau,\omega}$ nonsingular and $\mathbb{E}\|\phi_{i,\tau,\omega}^0\|^{2+\delta} < \infty$; $f_{\varepsilon_\tau|U,T,\mathbf{X}}(0) \in [c_f, C_f]$ locally Lipschitz at 0; K fixed folds with $\min_k n_k/n \geq c_K > 0$; the fold-uniform product-rate condition*

$$\max_k [r_{g,k}(r_{m,k} + r_{\beta,k}) + r_{\omega,k}(r_{m,k} + r_{\beta,k}) + (r_{m,k} + r_{\beta,k})^2] = o_p(n^{-1/2});$$

and fold-uniform score and bread consistency. Then the cross-fitted one-step estimator (5) satisfies

$$\sqrt{n}(\tilde{\beta}_{\tau,\omega}^{\text{CF}} - \beta_{\tau,0}) = S_{\tau,\omega}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_{i,\tau,\omega}^0 + o_p(1) \Rightarrow \mathcal{N}(0, \Sigma_{\tau,\omega}), \quad \Sigma_{\tau,\omega} = S_{\tau,\omega}^{-1} \mathbb{E}(\phi_{i,\tau,\omega}^0 \phi_{i,\tau,\omega}^{0\top}) S_{\tau,\omega}^{-\top},$$

with the variance subject-clustered. The default complete-event ($\omega \equiv 1$) and IPCW ($\omega = 1/G_{C,0}$) instances are both root- n for the same $\beta_{\tau,0}$, with no censoring-correction term (Lemma 1).

Practical inference. Theorem 1 is the formal inferential route, attached to the cross-fitted one-step estimator (denoted B-OS-CF). In applications we report the joint-sieve point estimator of (3) (denoted A-CV; complexity chosen by subject-level cross-validation) with a full-pipeline subject bootstrap, and use B-OS-CF as an orthogonal validation/sensitivity estimator. A complete root- n theory for the cross-validated joint estimator (a joint-sieve Bahadur expansion with random, CV-selected complexity) is not pursued here; the simulations of Section 5 show that the two estimators agree closely in bias and coverage, so the bootstrap inference reported for A-CV is empirically supported by the theory available for B-OS-CF. Proofs are in Supplement A.

5 Simulation studies

We summarize four studies; full designs, additional metrics, and audit/sensitivity analyses are in Supplement C. Throughout, $X_1 \sim \text{Bernoulli}(0.5)$, $X_2 \sim N(0, 1)$; onset is Cox–Weibull; competing death is Weibull; delayed entry gives $\approx 35\%$ truncation; $\approx 20\%$ loss to follow-up; $\approx 20\%$ death before onset; visits are noninformative ($\bar{m} \approx 4$). The log-marker is $L(u) = m_0(u, T) + \beta_{01}X_1 + \beta_{02}X_2 + (\sigma_0 + \sigma_1 X_1)\varepsilon$, giving the analytic truth $\beta_{\tau,1} = \beta_{01} + \sigma_1 z_\tau$, $z_\tau = \Phi^{-1}(\tau)$; Scenario 1 is homoscedastic ($\sigma_1 = 0$) and Scenario 2 ($\sigma_1 = 0.4$) makes the conditional density at the quantile depend on X_1 .

Finite-sample inference (Supplement Table C.1). For the complete-event A-CV and B-OS-CF estimators, bias of $\hat{\beta}_\tau$ is negligible at $\tau \in \{0.1, 0.5, 0.9\}$ and $n \leq 4000$, the bootstrap standard error tracks the empirical standard deviation, and subject-bootstrap coverage of $\beta_{\tau,1}$ is 0.918–0.986 across estimators, with mild undercoverage confined to lower-quartile cells where the conditional density is small. A-CV and B-OS-CF agree closely, consistent with the implicit orthogonalization of Section 3.2.

Censoring weights and orthogonality (Figure 1). The complete-event (CC) and IPCW estimators are both unbiased; CC is marginally tighter ($\text{ESD}_{\text{CC}}/\text{ESD}_{\text{IPCW}} = 0.95$). Estimating G_C adds no detectable first-order variance: the dispersion of the estimated-vs-oracle weight gap, $\text{sd}(\hat{\beta}_{\text{IPCW}} - \hat{\beta}_{\text{ORC}})$, decays at a log–log slope near -1 (i.e. $\approx n^{-1}$), faster than the $n^{-1/2}$ estimator scale, numerically supporting the second-order role of $\hat{G}_C$ (Lemma 1).

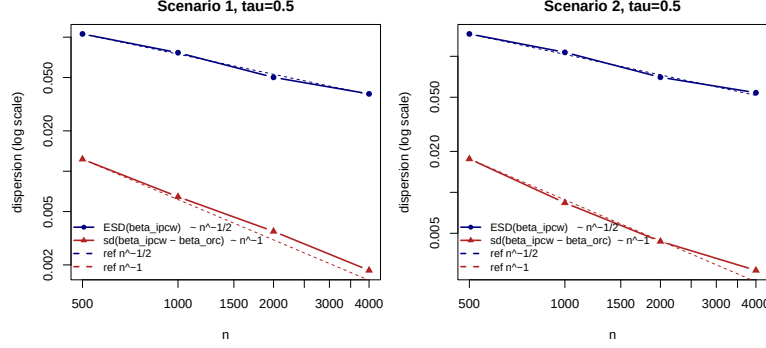


Figure 1: Censoring-weight diagnostic. The estimated-vs-oracle weight gap (red; fitted log–log slope near -1) shrinks faster than the estimator standard deviation (blue, slope $\approx -1/2$), numerically supporting the second-order ($\chi^C = 0$) censoring orthogonality of Lemma 1.

Distributional features beyond the mean. Under heteroscedasticity the estimated quantile-trajectory surfaces $\hat{m}_\tau(u, t)$ are non-parallel across τ , and $\hat{\beta}_\tau$ recovers the analytic τ -varying covariate effect; point estimation is stable under heavy-tailed (t_5) and right-skewed errors (Supplement C). This is the regime a conditional-mean model cannot describe and motivates the head-to-head comparison next.

5.1 Comparison with the conditional-mean estimator of Sun et al.

We compare against the *actual* estimator of Sun et al. [2025], run from their published code (the compiled profile partial-likelihood slope $\hat{\beta}$ and their kernel baseline $\hat{\mu}_0$), on a Sun-style survival/truncation/fixed-visit design with a Sun-favorable Gamma multiplicative-error marker, adapted to a binary group contrast. This is a *point-estimation* comparison ($R = 500$ replicates; Monte-Carlo bias, ESD and integrated mean-squared error (IMSE); no bootstrap). The conditional mean is reconstructed from the fitted quantile process two ways: the model-free quantile integral $\text{Q-INT} = \int_0^1 e^{Q_\tau} d\tau$ and the tail-stable log-location-scale retransformation $\text{Q-LN} = \exp\{Q_{0.5} + \frac{1}{2}\sigma^2\}$ (stronger than Sun’s multiplicative-mean assumption; we report both). Five regimes (Figure 2; Supplement Tables C.6–C.8 give the full numbers):

DGP1 (proportional mean, Sun-favorable Gamma). Both estimators are unbiased; the mean and median slopes coincide, and the direct quantile slope has smaller Monte-Carlo variability than the feasible Sun slope ($\sim 4\times$ smaller ESD). This is a point-estimation comparison, not an inferential-efficiency claim; an oracle- α decomposition (Supplement C) shows the difference is not due to estimating the onset coefficient. *DGP2 (proportional mean, covariate-driven dispersion).* Sun’s $\hat{\beta}_1$ stays unbiased for the mean, but only the quantile slope reveals that the covariate scales the spread ($\hat{\beta}_{.75} - \hat{\beta}_{.25} \approx 0.6$). *DGP3 (non-proportional induced mean contrast — the headline).* The mean log-contrast $\Delta(u, t)$ truly varies over (u, t) , so no constant proportional slope exists; Sun’s basic slope collapses it to a pseudo-average and his fully stratified Sec. 3.5 extension recovers the variation only at a large variance cost, whereas the quantile reconstruction tracks $\Delta(u, t)$ with several-fold lower IMSE (Figure 3). *DGP4 (tail-specific effect).* On the proportional mean target Sun is well matched; the quantile method’s value is *detection*—it reveals the effect is confined to a hidden upper-tail subgroup. *DGP5 (heavy-tailed error).* Under a t_5 log-error the conditional mean is undefined, so this is an estimand-validity illustration, not a performance contest: the quantile coefficients remain well-defined and stably estimated where the mean target is ill-posed.

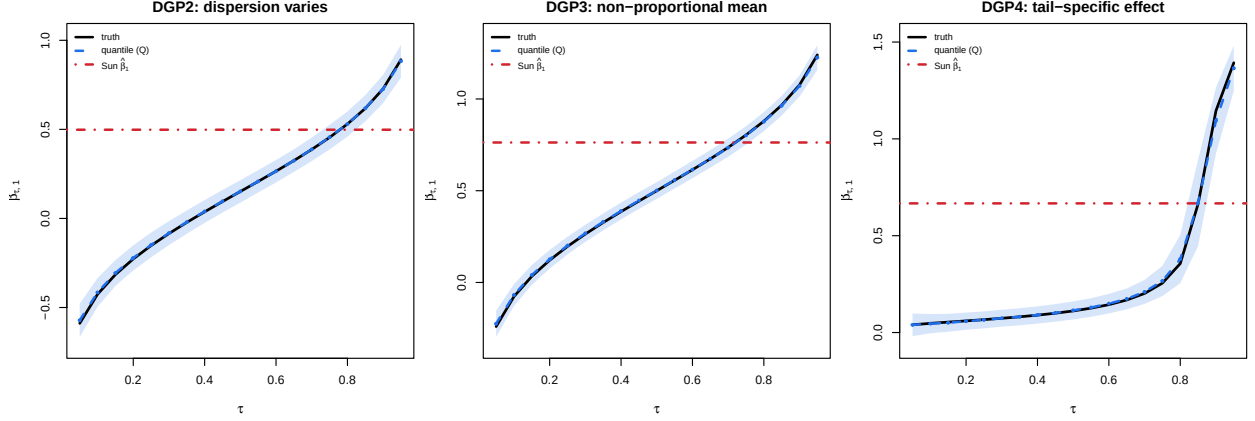


Figure 2: Covariate effect on the marker *distribution*: true $\beta_{\tau,1}$ (solid) vs. our quantile-regression estimate (dashed, ± 1 Monte-Carlo SD band) vs. Sun’s single mean slope $\hat{\beta}_1$ (flat dash-dot), $n = 1000$. DGP2 the mean is proportional yet the distributional effect varies with τ ; DGP3 the effect varies and the mean contrast is non-proportional; DGP4 the effect is confined to the upper tail. In each case the quantile estimate recovers the full curve while the scalar mean coefficient collapses it to one number.

6 Application: cortical thickness before amyloid positivity

Illustrative analysis on a synthetic ADNI-matched cohort; the real-data analysis on ADNI is in preparation and will replace this section.

We illustrate the method on a cohort matched to the Alzheimer’s Disease Neuroimaging Initiative (ADNI) design, in which a cortical-thickness biomarker is tracked before the landmark of amyloid positivity. Participants are amyloid-negative at staggered entry ages (delayed entry), are lost to follow-up, may die before becoming amyloid-positive (competing event), and are imaged at irregular visits; onset age T^0 is the age at first amyloid positivity. The covariates are an *APOE4* carrier indicator and sex. *Because access to the curated cohort is pending, the analysis below is run on a synthetic cohort generated to match the ADNI sampling design (entry, truncation, censoring, competing death, visit process) and serves to demonstrate the deliverables; it is not a substantive scientific finding.*

We report, on the estimable interior region: (i) the median trajectory surface $\hat{m}_{0.5}(u, t)$; (ii) the carrier effect $\hat{\beta}_{\tau,1}$ across τ with subject-bootstrap bands, which answers whether *APOE4* shifts the lower tail of thickness (accelerated thinning in the already-vulnerable) more than the median; and (iii) the induced mean contrast compared with the Sun et al. [2025] mean coefficient, showing what the mean summary misses (Figure 4). Sensitivity to the weighting (CC vs. IPCW), to the inferential route (A-CV vs. B-OS-CF), and to crossing (handled by rearrangement of predicted curves) is reported in Supplement D.

7 Discussion

We have introduced an event-aligned conditional quantile trajectory model for a biomarker observed before disease onset under left truncation, right censoring, competing death, and irregular visits, identified it without modeling the onset or censoring mechanisms, and given an orthogonal, cross-fitted one-step estimator with $\sqrt{n}$ inference. The conditional median effect $\beta_{0.5}$ is a location effect

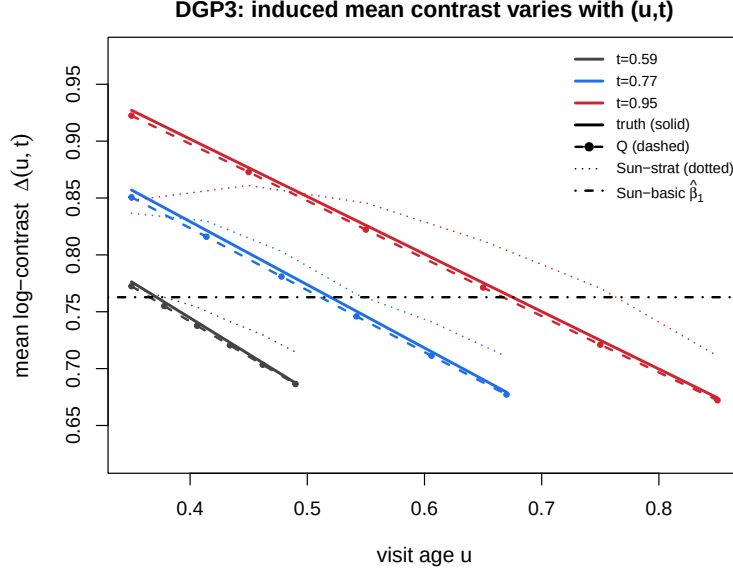


Figure 3: DGP3 mean log-contrast $\Delta(u, t) = \log\{\mathbb{E}[Y|x_1=1]/\mathbb{E}[Y|x_1=0]\}$ at three onset times, $n = 1000$: truth (solid), quantile reconstruction (dashed), Sun’s basic proportional model (flat dash-dot), and his stratified-baseline extension (dotted). The true contrast varies with (u, t) ; the basic model collapses it to a constant and the stratified extension is noisy, while the quantile reconstruction tracks it.

on the log-marker and need not equal the multiplicative mean coefficient of Sun et al. [2025] under heteroscedasticity. Against their actual estimator, the comparison is not that we estimate the mean better: when the proportional mean holds both are unbiased. The contribution is that the quantile family answers questions the mean model structurally cannot—covariate-driven dispersion, non-proportional induced mean contrasts, and tail-specific effects—and remains well-defined where heavy tails make the conditional mean ill-posed.

Robustness to the onset-time distribution (including non-proportional-hazards and accelerated-failure-time onset) is intrinsic to the conditional formulation, since the onset mechanism is never modeled. Genuine extensions include interval-censored or latent onset $T^0 \in (L, R]$, informative onset ascertainment, marginalization over T^0 , and informative visit processes (which reintroduce a visit weight). A complete root- n theory for the cross-validated joint-sieve estimator and a bootstrap consistency proof remain open and are natural next steps.

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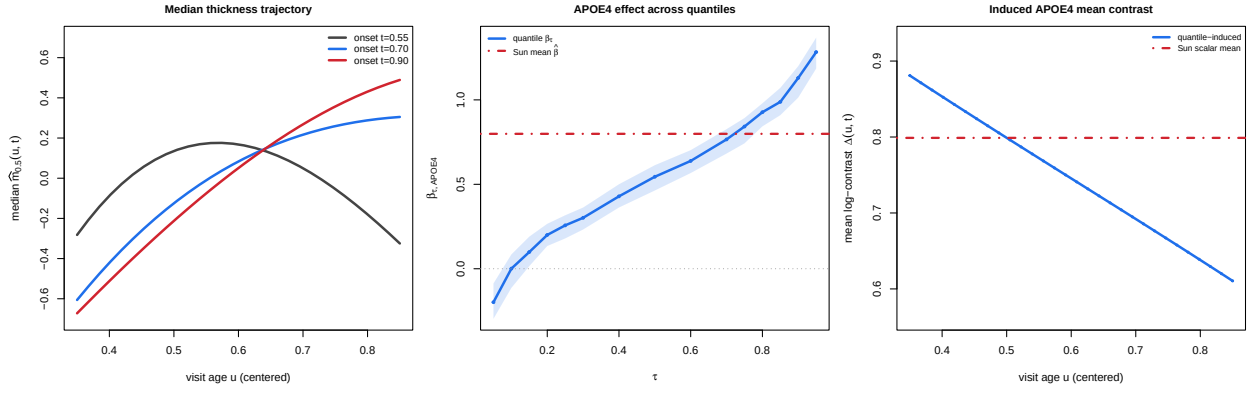


Figure 4: Illustrative analysis on a synthetic ADNI-matched cohort. Left: estimated median thickness trajectory $\hat{m}_{0.5}(u, t)$ over visit age u at three onset ages t . Center: $APOE4$ carrier effect $\hat{\beta}_{\tau, 1}$ across τ with 95% subject-bootstrap band; a non-flat curve indicates a distributional (not pure location) effect. Right: induced mean log-contrast from the quantile fit (curve) vs. the scalar conditional-mean coefficient (flat line). Synthetic data; for illustration of the outputs only.